ENGINE Pro - Version 3.1

Racing Systems Analysis - www.QUARTERjr.com

File: (untitled)
Note: Base case for ENGINE Pro

Input Engine Data

Number of Cylinders	Vee	8
Bore Diameter - inch		4.030
Stroke Length - inch		3.480
Rod Length - inch		5.850
Compression Ratio		12.9
Cam Type:	Normal Flat Tappet & Solid Lifter	
Intake Duration @ .050 inch - degree		264
Throttle CFM @ 1.5 inch Hg		750
Carburetor	Fuel Type:	Gasoline

Intake Manifold Type:	Common Plenum Style
Manifold Runner Style:	Curved Runner
Intake Manifold Flow Factor - %	96.0
Number of Intake Valves per Cylinder	1
Intake Valve Diameter - inch	2.050
Maximum Intake Port Flow - CFM	250.0
@ Test Pressure - inch H2O	28.0
@ Ref. Bore Diameter - inch	4.000

Compression Ratio Worksheet

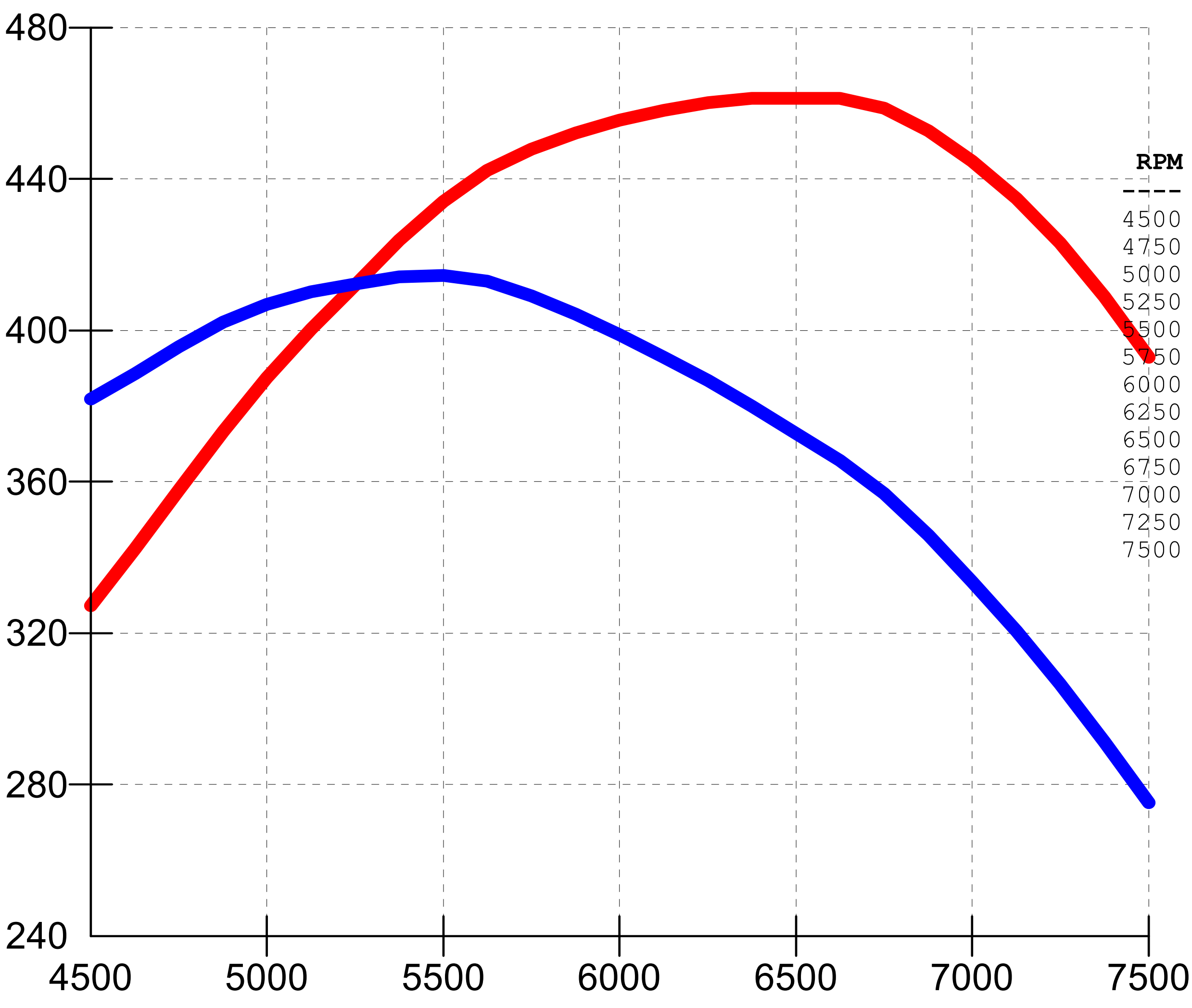
Combustion Chamber Volume - c.c.	62.0
Piston to Deck Height - inch	.015
Head Gasket Thickness - inch	.039
Piston Dome Volume - c.c.	12.0
Compression Ratio	12.9

Throttle CFM @ 1.5 inch Hg Worksheet

Butterfly Throttles:	Pri 4	Sec 0
Throttle Diameter - inch	1.688	.000
Venturi Diameter - inch	1.375	.000
Throttle CFM @ 1.5 inch Hg		730

Estimated Performance for 355.1 CID Engine

				SI Units	
Peak HP	461	Peak Torque - ft lbs	415		
RPM @ Peak HP	6650	RPM @ Peak Torque	5450	Displacement - liter	5.82
Peak HP/CID	1.30	Peak Torque/CID	1.17	Peak Power - kW	344
Shift RPM	7200	Redline RPM	8350	Peak Torque - Nm	562



RPM	HP	TQ
4500	327	382
4750	358	396
5000	387	407
5250	412	412
5500	434	415
5750	448	409
6000	456	399
6250	460	387
6500	461	373
6750	459	357
7000	445	334
7250	423	306
7500	393	275

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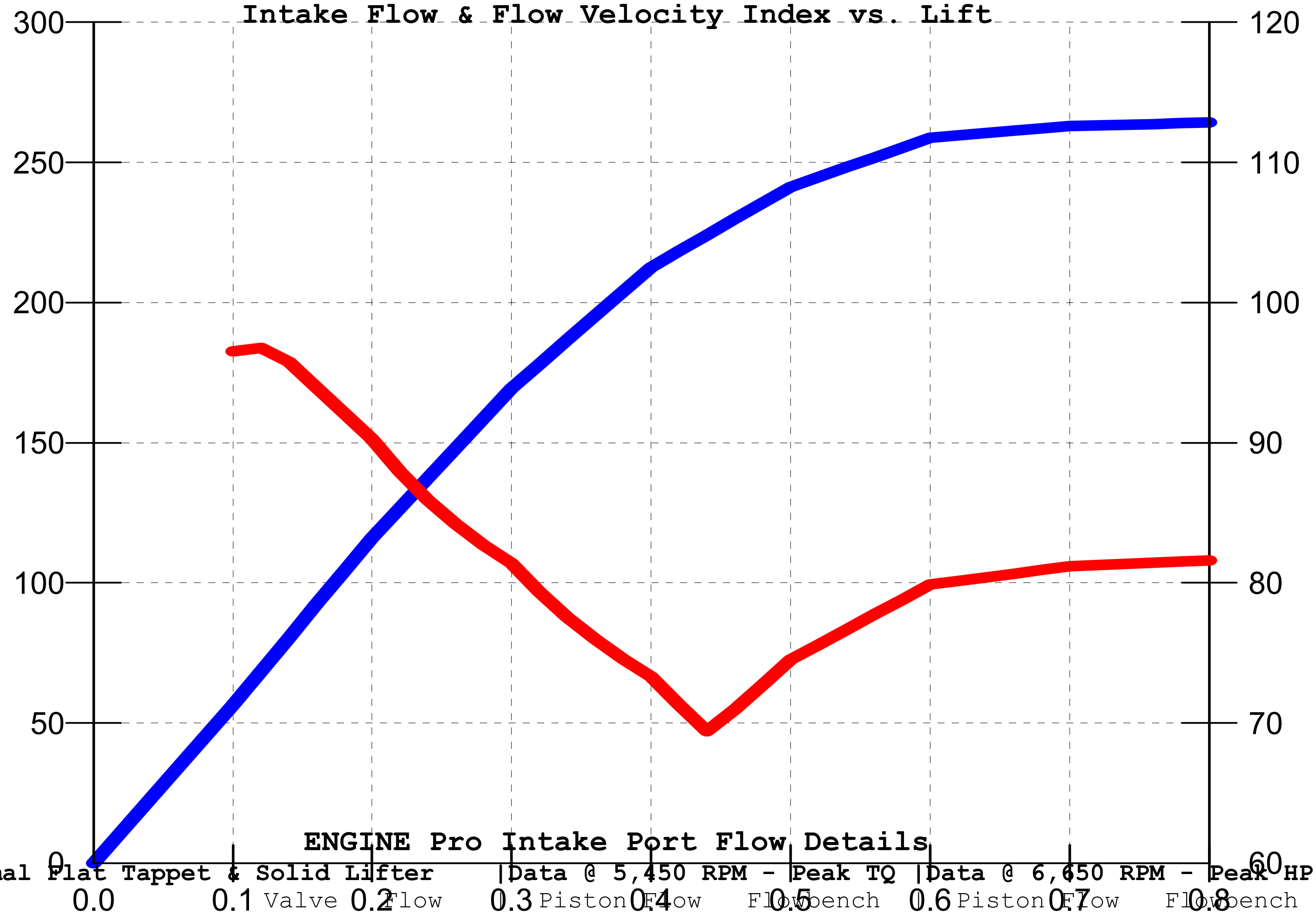
Intake Port Flowbench Data @ 28.0 in H2O Worksheet

Number of Intake Valves per Cylinder	1
Intake Valve Diameter - inch	2.050
Valve Seat Throat Diameter - inch	1.794
Valve Seat Throat Percentage - %	87.5
Valve Seat Angle - degree	45.0
Valve Seat Width - inch	.080
Valve Stem Diameter - inch	.344
Maximum Intake Valve Lift - inch	**.550**
Flow @ Maximum Valve Lift - CFM	250.0
Valve Seat Throat Area - sq inch	2.435
Intake Flow Velocity - ft/sec	246.4
Intake Flow Flux - CFM/sq inch	102.7
Flow Velocity Index - %	77.2

Lift Inch	Flow CFM	Area sq in	Velocity ft/sec	Flow Flux CFM/sq in	Flow Vel Index-%
.100	56.6	0.441	308.0	128.3	96.5
.200	116.0	0.966	288.2	120.1	90.3
.300	169.4	1.565	259.8	108.2	81.4
.400	212.6	2.180	234.1	97.5	73.3
.500	241.3	2.435	237.8	99.1	74.5
.600	258.7	2.435	255.0	106.2	79.9
.700	262.9	2.435	259.1	108.0	81.2
.800	264.2	2.435	260.4	108.5	81.6

Intake Flow & Flow Velocity Index vs. Lift



ENGINE Pro Intake Port Flow Details

Normal Flat Tappet & Solid Lifter

Event	deg ATDC	Valve Lift inch	Flow Area sq in		Data @ 5,450 RPM - Peak TQ: Piston Speed FPM	Flow Demand CFM	Flowbench Vel FPS	Test inH2O		Data @ 6,650 RPM - Peak HP: Piston Speed FPM	Flow Demand CFM	Flowbench Vel FPS	Test inH2O
IVO @ .050"	-27	0.050	0.218	|	-2857	0	--	--	|	-3486	0	--	--
TDC	0	0.163	0.757	|	0	-5	-15	--	|	0	-7	-22	--
30 deg ATDC	30	0.333	1.765	|	3129	228	311	42	|	3818	266	361	54
60 deg ATDC	60	0.471	2.435	|	4962	349	344	57	|	6054	402	396	72
Max Piston FPM	**74.6**	**0.515**	**2.435**	|	5182	370	365	58	|	6323	427	421	74
90 degree	90	0.541	2.435	|	4965	369	363	56	|	6059	427	421	72
ILC - Max Lift	**105**	**0.550**	**2.435**	|	4411	347	342	50	|	5382	407	401	66
120 deg ATDC	120	0.541	2.435	|	3638	308	303	41	|	4439	367	362	56
150 deg ATDC	150	0.471	2.435	|	1836	192	189	20	|	2240	240	236	29
BDC	180	0.333	1.765	|	0	46	63	2	|	0	68	93	4
25 deg ABDC	205	0.190	0.909	|	-1528	-82	-216	--	|	-1865	-88	-233	--
IVC @ .050"	237	0.050	0.218	|	-3468	0	--	--	|	-4231	0	--	--

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ENGINE Pro Mechanical Details

Piston Speed Summary - FPM

Rating	RPM	Avg	Max*
Peak TQ	5450	3161	5181
Peak HP	6650	3857	6322
Shift	7200	4176	6845
Redline	8350	4843	7939

*Maximum Piston Speed occurs @ 74.6 degree after TDC

Est. Cranking Compression - psig 230

Geometric Data Summary

Bore to Stroke Ratio	1.16
Rod to Stroke Ratio	1.68
Piston to Head/Rod Length	0.0092
Intake Throat/Bore Area Ratio	0.191
Intake Valve Lift/Diameter Ratio	0.268

Data @ 6,650 RPM - Peak HP

deg ATDC	depth inch	Piston Speeds FPM	FPS	g's accel
5	0.009	685	11	2818
15	0.077	2020	34	2679
30	0.298	3818	64	2233
45	0.640	5206	87	1561
60	1.067	6054	101	768
74.6	**1.524**	**6323**	**105**	**1**
80	1.694	6289	105	-257
85	1.851	6199	103	-479
90	2.005	6059	101	-681
105	2.437	5382	90	-1149
120	2.807	4439	74	-1417
135	3.101	3362	56	-1530
150	3.312	2240	37	-1553
165	3.438	1116	19	-1543
180	3.480	0	0	-1535

Data @ 6,650 RPM - Peak HP

deg ATDC	depth inch	Piston Speeds FPM	FPS	g's accel
5	0.009	685	11	2818
15	0.077	2020	34	2679
30	0.298	3818	64	2233
45	0.640	5206	87	1561
60	1.067	6054	101	768
74.6	**1.524**	**6323**	**105**	**1**
80	1.694	6289	105	-257
85	1.851	6199	103	-479
90	2.005	6059	101	-681
105	2.437	5382	90	-1149
120	2.807	4439	74	-1417
135	3.101	3362	56	-1530
150	3.312	2240	37	-1553
165	3.438	1116	19	-1543
180	3.480	0	0	-1535

Data @ 7,200 RPM - Shift

deg ATDC	depth inch	Piston Speeds FPM	FPS	g's accel
5	0.009	741	12	3304
15	0.077	2187	36	3141
30	0.298	4134	69	2617
45	0.640	5636	94	1830
60	1.067	6555	109	901
74.6	**1.524**	**6846**	**114**	**1**
80	1.694	6809	113	-302
85	1.851	6712	112	-562
90	2.005	6560	109	-798
105	2.437	5827	97	-1347
120	2.807	4806	80	-1661
135	3.101	3641	61	-1794
150	3.312	2425	40	-1820
165	3.438	1209	20	-1809
180	3.480	0	0	-1800

Data @ 8,350 RPM - Redline

deg ATDC	depth inch	Piston Speeds FPM	FPS	g's accel
5	0.009	860	14	4443
15	0.077	2536	42	4224
30	0.298	4794	80	3520
45	0.640	6536	109	2461
60	1.067	7602	127	1211
74.6	**1.524**	**7939**	**132**	**1**
80	1.694	7896	132	-406
85	1.851	7784	130	-756
90	2.005	7607	127	-1073
105	2.437	6758	113	-1812
120	2.807	5574	93	-2234
135	3.101	4222	70	-2412
150	3.312	2813	47	-2448
165	3.438	1402	23	-2432
180	3.480	0	0	-2421

ENGINE Pro Recommendations

Intake System:

Intake Valve Lift - inch	.580
Minimum Flow Area - sq inch	2.55
Total Intake Track Length - inch	14.75
Maximum Flow Area - sq inch	3.50
Total Intake Track Volume - c.c.	680
Plenum Volume - cubic inch	355

Camshaft:

Lobe Separation Angle - deg	108
Intake Lobe Centerline - deg	105
Exhaust Duration @ .050 inch - deg	278

Exhaust Port:

Exhaust Flow - CFM @ 28.0 inch H2O, %Intake @ 4.000 inch Ref. Bore Diameter	160=64%
Exhaust Valve Diameter - inch	1.50-1.54
Exhaust Valve Lift - inch	.520
Minimum Flow Area - sq inch	1.42
Maximum Flow Area - sq inch	2.04

Exhaust System:

Primary Tube Length - inch	36.0
Primary Tube Diameter - inch	1.750
Collector Diameter - inch	3.25